Prime-length necklaces

The mathematics in one view

A **readout** is the clockwise sequence of bead colors from a marked start. A **ring** identifies readouts that differ by rotation only. Reflections are not identified: keep rings face up. A ring with n positions can have fewer than n distinct readouts when its coloring repeats.

Period divides length. The number d of distinct readouts is the smallest positive rotation that restores the coloring, and d divides n. Equivalently the ring consists of n/d repeats of a block of length d. Constant rings have d=1. For prime length p, the only possible values are 1 and p; every ring using at least two colors has p distinct readouts.

Counting and Fermat. With $c \geq 1$ colors and p prime, there are c^p marked readouts, of which c are constant. The remaining c^p-c readouts partition into rotation families of size p. Thus p divides c^p-c, and the number of rings is $c+(c^p-c)/p$. This is the divisibility form of Fermat's little theorem, with a concrete counting proof.

Limits. Do not divide all readouts by the ring length: constant and periodic rings have smaller families. For a composite length, a mixed ring can repeat early; the prime formula need not count correctly. The congruence for a particular color count is not a converse primality test.

What each group is doing

K-1: Make rings, match by turning, and compare the number of different starts on small examples. Recognizing A/B, counting to 6 and moving a start arrow are enough. Physical sorting and a complete short case list are genuine mathematical outcomes.

Grades 2-3: Partition the supplied cards for lengths 3 and 5, contrast a length-4 early repeat and six-bead rings with two, three and six readouts, and conjecture or explain why mixed five-bead rings have five views. Counting to 32 and equal-sized grouping are useful; reading all cards aloud is unnecessary.

Grades 4-5: Compare six, seven and nine beads, prove the prime-length orbit claim, count rings with three or c colors, and derive divisibility. Product counting and division are needed for the final formula. A class that establishes only the length-5 case has not yet proved the prime-length theorem.

Materials and preparation

Five pair-kits of six A and six B counters. The printed boards are the mats; to turn one ring against another, cut a spare page of four- or five-bead boards into single-board mats. A sticky note cut to an arrow is the movable start. Use flat face-up mats rather than jewelry that can flip. Print one extra eight-card page per table and one 32-card page for each older table, and bring scissors for the cards; older children can cut the card rectangles themselves. A no-cut option is to mark each family in one color on the supplied sheet. Add a third counter color only for upper P7.

Returning children may recognize rotation from Week 4. The new destination is equal-sized families at prime length and their divisibility consequence, not a new claim about rotation itself. Unpiloted.

Launch and a flexible hour

0-5, after the opening run-around: Launch. Let children place counters on a three-position mat. Use the printed AAB example: keep the ring fixed and move the start clockwise, obtaining AAB, ABA, BAA. These are three readouts of one ring. Keep reading clockwise and never turn the mat face down.

5-20: Work on the three-bead rings and eight-card sort. **20-38:** Bring a ring to another pair and ask whether it matches theirs by turning, then compare the four-bead case or sort five-bead cards cooperatively. **38 to the end:** Share a group-size claim and its explanation. Minutes count from the launch; the tables have about 35-40 minutes. K-1 can stay with three and four beads on a first visit, with five and six beads for a return; upper pupils can take the prime argument only after they own the smaller examples.

Three-bead key: K P1-2; middle and upper P1-2

One ring representative	All different readouts	Printed card numbers
AAA	AAA	1
AAB	AAB, ABA, BAA	2,3,5
ABB	ABB, BBA, BAB	4,6,7
BBB	BBB	8

There are four rings, with family sizes 1,3,3,1. The group sizes are not all the same. All eight words are accounted for, so the list is complete. Dividing 8 by 3 would wrongly treat AAA and BBB as three-card families. The correct count is 2+(8-2)/3=4.

K-1 answer key, P3-6

P3. Exactly two rings have two As and two Bs: AABB (the As adjacent) and ABAB (the As opposite). Each pair of A positions is either adjacent or opposite, exhausting the possibilities.

P4. One readout: AAAA or BBBB. Two: ABAB, with readouts ABAB and BABA. Four: AAAB, with AAAB,AABA,ABAA,BAAA; AABB and ABBB also work. Exactly three is impossible; a return must fit evenly into four positions (proof on page 4).

P5. Exactly two rings: AABBB and ABABB. The two A beads are adjacent or separated, up to turning; on five positions there is only one nonadjacent separation. Each has five different readouts. AABBB gives AABBB,ABBBA,BBBAA,BBAAB,BAABB; ABABB gives ABABB,BABBA,ABBAB,BBABA,BABAB.

P6 (also Grades 2-3 P7). Examples using both colors: ABABAB has two readouts; AABAAB has three; AAAAAB has six. The first repeats AB, the second repeats AAB, and the last has one B whose six positions distinguish all starts. No exhaustive list of six-bead rings is requested.

Grades 2-3 answer key, P3-7

P3. The possible numbers are 1,2,4, with examples AAAA, ABAB, AAAB. Three is impossible. A repeating block must fit an integer number of times around the ring; the block-length proof is on page 4. A physical alternative: any repeat is a match after 1, 2 or 3 steps. After 1 or 3 steps, following the match around reaches all four positions, so the ring is one color (one readout). After 2 steps, opposite beads agree, so there are at most two readouts. Three never happens.

P4. There are eight five-bead rings: two singleton families and six families of five. These are the exact groups for the printed 32 cards, whose numbering is in alphabetical order.

Representative	Card numbers in its rotation family	Size
AAAAA	1	1
AAAAB	2,3,5,9,17	5
AAABB	4,7,13,18,25	5
AABAB	6,10,11,19,21	5
AABBB	8,15,20,26,29	5
ABABB	12,14,22,23,27	5
ABBBB	16,24,28,30,31	5
BBBBB	32	1

P5. No mixed five-bead ring can have fewer than five readouts. If a rotation by 1,2,3 or 4 places restores the ring, repeatedly moving by that amount visits all five positions. All those positions must carry the same color, contradicting the ring being mixed. For an accessible demonstration, label positions 0-4: steps of 2 visit 0,2,4,1,3,0; steps of 3 reverse that order. Steps of 1 or 4 also visit every position.

P6. The two constant words do not belong to five-card families, so 32/5 is invalid. Count them separately: 2+(32-2)/5=2+6=8.

P7. The K-1 P6 examples on page 2: ABABAB (two readouts), AABAAB (three), AAAAAB (six). Two, three and six all divide six; four and five never occur. This is the contrast to P5, where every mixed five-bead ring has five readouts.

Hints, only when needed

If children count a ring twice, build both on single-board mats cut from a spare board page, keep one fixed and turn the other through every position. If a family has a repeated card, ask whether the whole readout changed, not merely the arrow. For completeness, sort by the number of A beads. For the 32-card page, assign different first-two-letter branches to pairs, then merge rotation matches. Give the cards rather than ask a child to manufacture all 32 words.

When a child says “every mixed ring has as many views as beads,” test ABAB before revealing the prime rule. This makes the hypothesis earn its place.

Grades 4-5 answer key, P3-7

P3. Sizes 1,2,4 are possible; size 3 is not. The six length-4 rings are represented by AAAA,BBBB (size 1), ABAB (size 2), and AAAB,AABB,ABBB (size 4). A repeated shorter block, here AB twice, lets a mixed ring return early.

P4. Use the complete 32-card key on page 3. The result is eight rings: two of size 1 and six of size 5, so 2+(32-2)/5=8. Four admits a nonconstant block of length 2 repeated twice; five has no proper divisor except 1, whose block would be constant.

P5. Six beads: 2, 3 or 6 readouts (ABABAB, AABAAB, AAAAAB). Seven beads: only 7. Nine beads: 3 or 9 (AABAABAAB, AAAAABBBB). A ring using both colors never has one readout, and the number always divides the length; seven, a prime, leaves only itself. This is the pattern P6 asks children to state and prove.

P6. Exactly p distinct readouts for a prime-length p ring using at least two colors. If rotation by k positions, $0<k<p$, restored the coloring, the positions 0,k,2k,... modulo p would all have the same color. These visit all p positions: a repetition before p steps would give p dividing mk for $0<m<p$, impossible since p is prime. The ring would be constant. Thus every nonzero rotation changes it and all p starts differ.

P7. With three colors and length 5, there are $3^5=243$ readouts. Three are constant; the other 240 form 48 families of five. Total: **51 rings**. Generally the count is $c+(c^p-c)/p$, because each of the p positions independently has c choices. The nonconstant readouts partition into families of p, proving p divides c^p-c.

Why the smallest period divides the length

Let d be the least positive rotation that restores an n-position ring; n itself always does. Write $n=qd+r$ with $0\le r<d$. A rotation by qd restores it, and so does a full turn n; consequently their difference r also restores it. Minimality forces r=0, so d divides n. Starts separated by d give the same word; starts with different remainders modulo d cannot agree, or a smaller positive difference would restore the ring. Hence there are exactly d distinct readouts.

This is the missing justification behind “the block must fit.” It proves the composite obstruction as well as the prime case. A child’s complete physical sort proves the particular finite count; the period argument explains why the count generalizes.

Optional extension: why face up matters

Compare the six-bead rings AABABB and AABBAB. They match after a reflection but not by rotation. The rotations of AABABB are AABABB, ABABBA, BABBAA, ABBAAB, BBAABA, BAABAB; none is AABBAB. This gives a small concrete reason that allowing flips would change the objects being counted.

Primary source

University of Oregon, Math 391 (Abstract Algebra), *Worksheet 6*, 12 October 2020, problems 1-6, one page: the prime-length necklace proof of Fermat’s little theorem and the contrast with length 4. Read the original instructor worksheet. This is a source for the mathematical proof, not evidence of K-5 classroom use. The child tasks, card partitions and small cases above were independently checked.

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

The movable-card method already exists: use the supplied eight- and 32-card sheets, cut into rectangles before use if desired, and group the cards physically; no handwritten reproduction of the catalog is required. Keep one adult at each table, with partners sharing and checking rotation families. For another visit, K-1 can compare four-bead periods (P3-4), then the five-/six-bead examples (P5-6). Grades 2-3 can sort the supplied five-bead cards cooperatively (P4), justify the count (P5-6) and find six-bead rings with two, three and six readouts (P7). Grades 4-5 can compare six, seven and nine beads (P5), then return to the prime-period theorem and general color count (P6-7), using the saved families as examples. Keep rings face up.